ARCTIC TRAIL CAM — OFF-GRID CABIN & PRIVATE LAND SECURITY

Movement Detection for Remote Cabins, Hunting Grounds & Private Forest Lands Far From Internet

Designed for Harsh Finnish Arctic Conditions (-40°C to +35°C Year-Round) • 120+ Day Battery Life • Zero Radio Waves • $542 Off-the-Shelf

FAR FROM INTERNET OR CELL

Zero Wi-Fi or Cell Needed

- Built for deep wilderness cabins and private lands miles from the nearest cell tower or power grid.
- Sends crisp 1080p photos via 3mm moss-buried glass fiber or our 200m–20km all-weather sky-bounce troposcatter relay over hills.

GREAT FOR HUNTERS & OWNERS

Trails, Driveways & Lakeshores

- Ideal for cabin owners, private landowners, and hunters.
- Spots people, vehicles, boats, and wildlife day or night.
- Low-profile solid forest-green enclosure, zero red night glow, and zero radio signals at the tree.

HARSH FINNISH ARCTIC (-40°C)

Built for Year-Round Weather

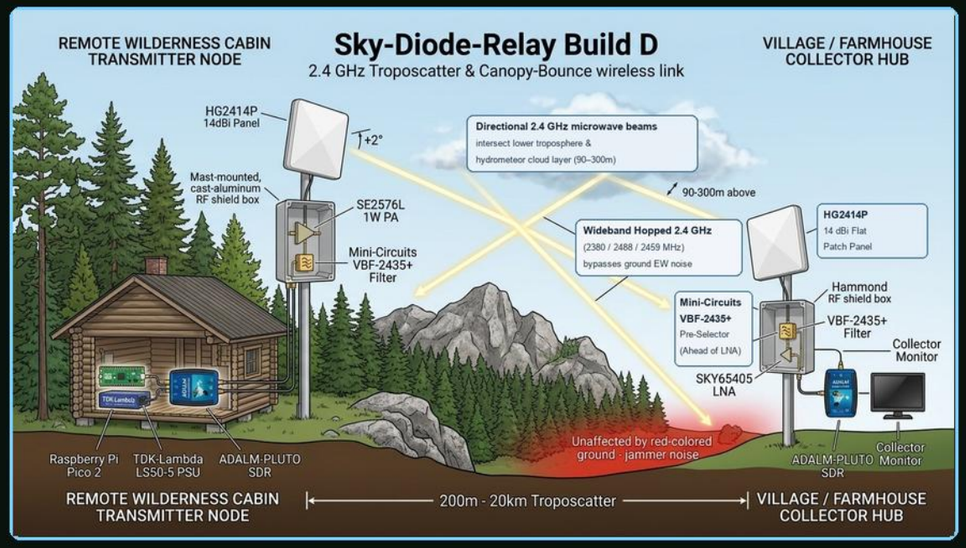
- Engineered for freezing Finnish winters (-40°C), snow, sleet, lake fog, and autumn rain.
- Battery sits buried under warm forest moss; automatic 4-second window heater clears frost from the camera lens in deep cold.

9–12+ MO BATTERY & AI

10 Photos/Mo (or 120d @ 10/Day)

- Main AI computer stays 100% turned off (0 Watts) while a tiny 0.004W sensor watches the trail.
- Wakes in 6s on movement, confirms target with AI, sends photo & shuts off.
- 9–12+ months at 10 photos/month (or 120+ days at 10 photos/day)!





INSIDE THIS OVERVIEW: [Page 2] Tree Camera in Context • [Page 3] Inside the 3 Boxes ($542) • [Page 4] Cabin & Private Land Setups • [Page 5] 200m–20km Sky-Diode-Relay (Build D) • [Page 6] 58-Photo Benchtop Build

How the 2-Piece Forest-Green Camera Blends into a Pine Tree

Splitting the tiny 'Eye' from the 'Brain & 120-Day Battery' keeps the tree unit tiny and protects the battery under warm moss in -40°C winter.



ZONE 1 (TOP — 2m HIGH)

Fist-Sized Camera & Radar Pod

- Smaller than a deck of cards (11 x 6 cm), finished in matte forest green for trees.
- Only ONE small glass window (with an internal foam ring to block night glare).
- Hidden radar sees through the solid plastic box without any extra holes.

ZONE 2 (MIDDLE — 2m CORD)

Matte Forest-Green Tree Cord

- A single thin 6mm cord in a matte forest-green protective sleeve.
- Tucks neatly inside a natural groove in the pine bark down to the roots.
- Carries power up and video down.

ZONE 3 (BOTTOM — UNDER MOSS)

Moss-Buried Brain & 120d Battery

- Solid forest-green waterproof box buried under forest moss at the tree roots.
- Natural ground insulation keeps the 12V battery warm during -40°C Finnish winters for 120+ days of year-round operation.
- Houses the AI computer & light sender.

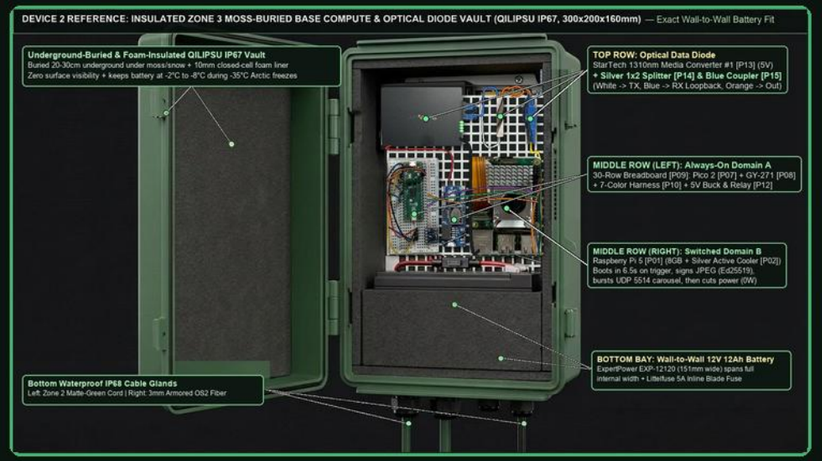
How It Achieves 120+ Days of Arctic Winter Battery Life (The 6.8-Second Wake-Up Cycle):

1. DEEP WINTER SLEEP (0.004 Watts): The main computer is completely turned off. Only a tiny sensor chip watches for cars, gear, motion, or footsteps.
2. INSTANT WAKE-UP (6.8 Seconds): When movement is detected, it flips on power to the Raspberry Pi 5, snaps a 1080p photo, and runs AI detection.
3. ONE-WAY FLASH OF LIGHT & POWER-OFF: The photo is flashed down the glass fiber to the cabin, and the computer immediately shuts back off to 0 Watts.

Three Modular Units Built 100% from Off-the-Shelf Parts ($542 Total)

Engineered for harsh Finnish year-round conditions — standard off-the-shelf gear in clean, solid forest-green outdoor enclosures.







UNIT 1: EYE & RADAR POD ($59)

Inside the Tree Camera Box

- Top Half: Wide-angle night-vision camera surrounded by a black foam ring so the invisible 940nm night LEDs never glare back into the camera lens.
- Arctic Window De-Icer: Tiny heater clears frost from the lens in 4s at -40°C.
- Bottom Half: 24 GHz motion radar sits behind solid, uncut matte-green plastic — sealed against snow, ice, and rain.

UNIT 2: BURIED BASE VAULT ($402)

Inside the Moss-Buried Brain Box

- Bottom Row: Standard 12V battery fits snugly wall-to-wall; delivers 120+ days of unattended year-round battery life.
- Middle Left: Tiny 'Gatekeeper' chip that sips 0.004W while waiting for movement.
- Middle Right: Raspberry Pi 5 AI computer (turned off 99.9% of the time).
- Top Row: Optical converter + yellow 1-to-2 light splitter feeding the glass fiber.

UNIT 3: CABIN RECEIVER ($81)

Inside the Cabin or Woods Vault

- Option 3A (Cabin Laptop): The glass fiber from the woods plugs into a small box next to your cabin laptop — displaying instant movement alerts with zero internet.
- Option 3B (Buried Woods 'Black Box'): For hunting land with no cabin nearby, saves photos onto a hidden underground drive 200m back in the forest and beams them 200m–20km over hills to a farmhouse.

How Alerts Reach You: Direct Hidden Fiber vs. Remote Woods Vault

Works whether your cabin is 50 meters from the driveway or you are monitoring remote hunting land kilometers from the nearest road.

SCENARIO A: CABIN NEARBY (30m to 10km — Direct Moss-Buried Fiber)



SCENARIO B: NO CABIN NEARBY / 200m–20km TROPOSCATTER / WATER CROSSING



SCENARIO A (LEFT PANEL)

Cabin Nearby (30m to 10km)

- Unroll the thin 3mm steel-armored glass fiber cable from the trail tree to your cabin and press it 2 cm under the moss (no trench digging needed!).
- Glass fiber runs up to 10 kilometers through freezing snow or wet bogs with zero signal loss and zero radio waves.
- Plugs straight into your cabin laptop.

SCENARIO B (RIGHT PANEL)

Remote Private Land / No Cabin

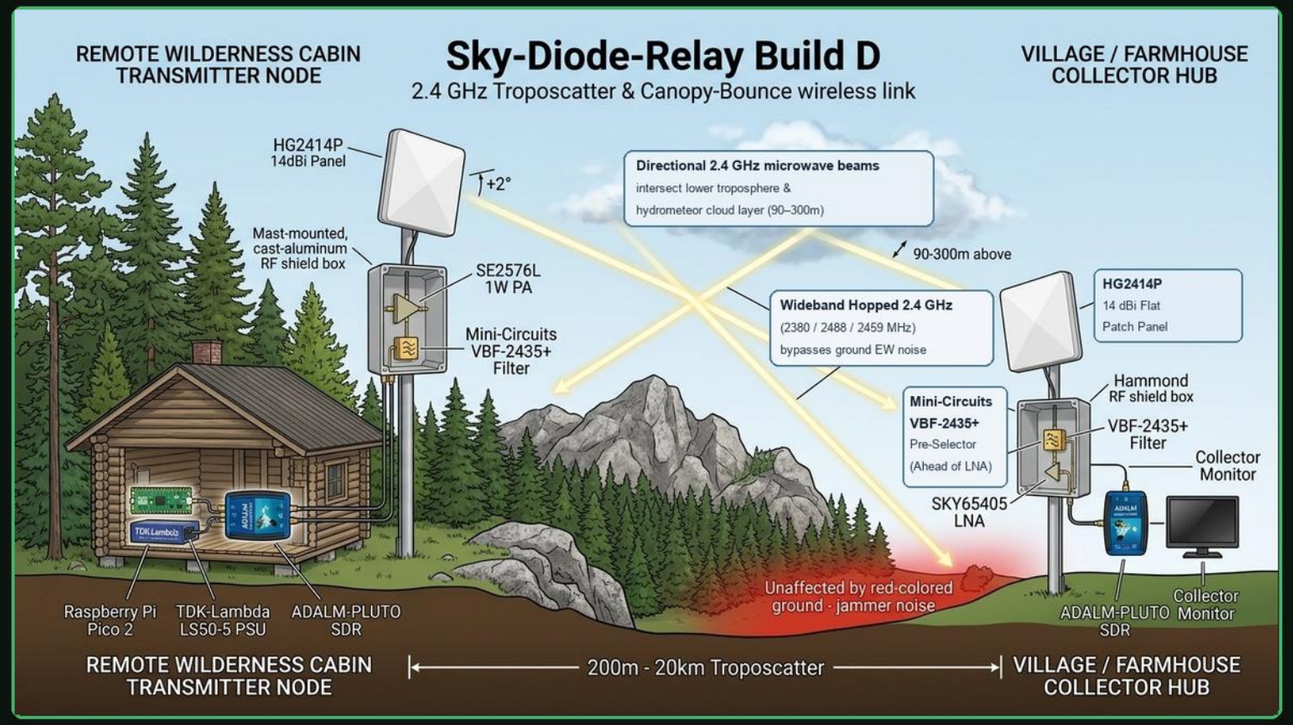
- Monitoring remote hunting grounds or a distant cabin 200m–20km over a hill?
- A 200m glass fiber hides a second solid forest-green vault deep in the woods — so even if someone spots or damages the trail camera, the photos are safe 200m away!
- From the woods, relays alerts up to 20km over hills via Sky-Diode-Relay Build D.

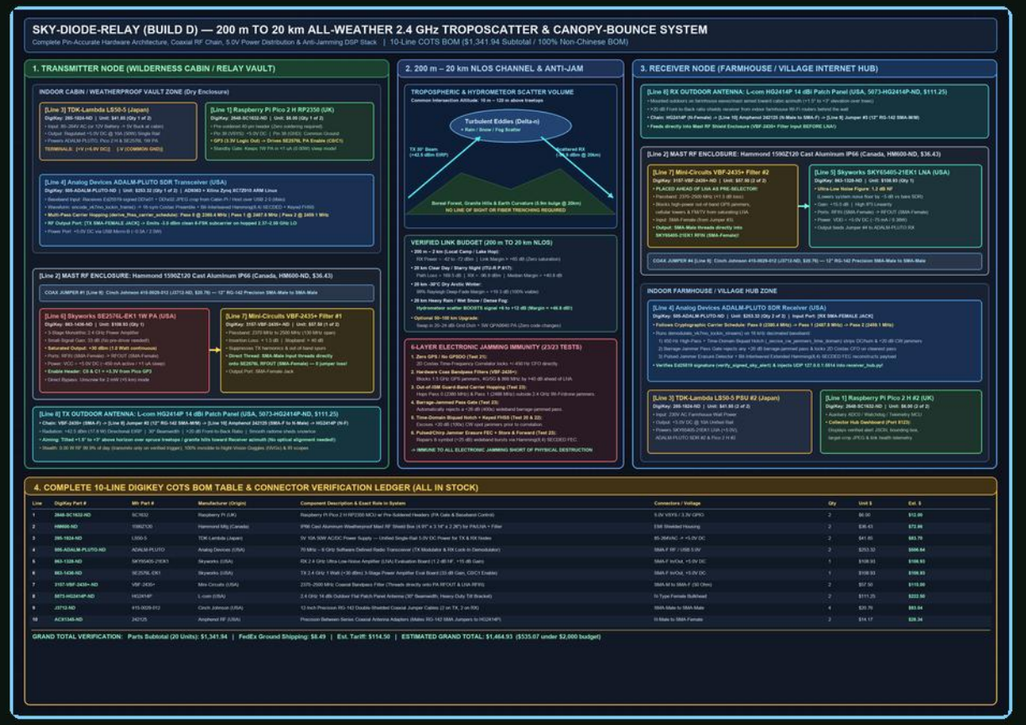
Built for Complete Off-Grid Independence & Tamper-Proof Security:

- Zero Monthly Fees, Zero Cloud Servers, Zero Cell Towers: Works in the deepest wilderness of eastern or northern Finland.
- Strictly One-Way Link: Data only travels from the camera node to the collector. Nothing can ever travel backwards to compromise the camera.
- Locally Replaceable 12V Battery: Uses a standard €27 12V 12Ah battery from any local auto/marine store (Motonet / Biltema) for 120+ days per charge.

Over-the-Hill 200 m – 20 km All-Weather Troposcatter & Canopy-Bounce Link

Ordered COTS Hardware ($1,341.94 Parts Subtotal) • 35–45 Min Zero-Solder Assembly • 6-Layer Anti-Jamming





ORDERED BUILD D (200m – 20km NLOS)

All-Weather 2.4 GHz Troposcatter

- Bounces 2.4 GHz VK7MO lock-in frames (+55.8 dB gain) off tropospheric turbulence, rain/snow clouds & forest canopies.
- Works 24/7 in clear -30°C night, 24h summer sun, fog, rain & heavy snow (+14.1 dB to +62.3 dB net SNR).
- 35–45 min zero-solder SMA coax build!

6-LAYER ELECTRONIC ANTI-JAMMING

Out-of-ISM Guard-Band Hopping

- Hops across 2380 / 2488 / 2459 MHz: 2 of 3 passes sit in guard bands outside 2.4 GHz Wi-Fi & drone barrage jammers.
- Mini-Circuits VBF-2435+ coaxial filter ahead of Skyworks 1.2 dB NF LNA.
- MAD barrage-jammer pass rejection + 2D Costas lock-in (zero GPS needed).

ORDERED 10-LINE DIGIKEY BOM

20 Units ($1,341.94) + Water FSO

- 2x ADALM-PLUTO SDRs + Skyworks 1W PA (SE2576L-EK1) + 1.2 dB LNA (SKY65405).
- 2x Mini-Circuits VBF-2435+ Filters + 2x L-com HG2414P 14 dBi Flat Panels.
- 2x TDK-Lambda 5V 10A PSUs + 2x Hammond IP66 RF Boxes + 4x RG-142 SMA Jumpers.
- Also supports 5m–100m water laser FSO.

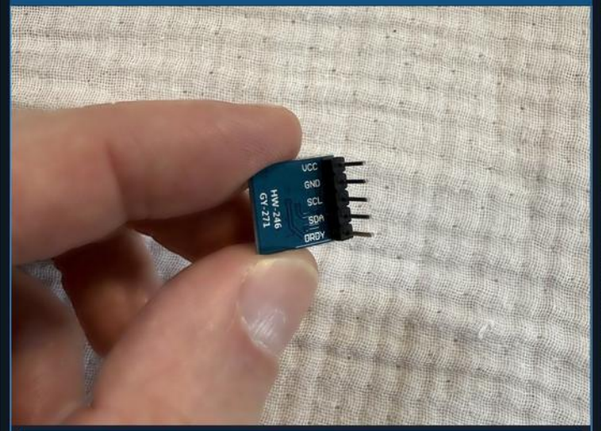
Photographed 58-Step Benchtop Assembly & Vibration-Locked COTS Design

Every jumper pin, 5-pin compass press-fit, CSI camera ribbon latch, and cold-boot power domain verified on physical hardware.

HOLE-COUNTING LANDMARK GUIDE: 4 WIRES IN COL f (3 EMPTY BOTTOM) vs. 5-PIN COMPASS IN COL j (2 EMPTY BOTTOM)

Because the GY-271 Compass header has 5 pins (vs. 4 wires), leave 3 empty bottom holes in Col f and 2 empty bottom holes in Col j!

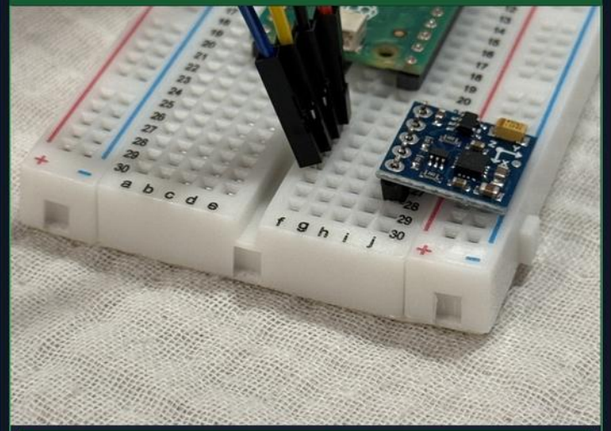
LEFT (PART IN HAND): 5-PIN GY-271 COMPASS MODULE (VCC AT TOP, DRDY AT BOTTOM)



WHY THE 5-PIN STRIP REACHES 1 HOLE LOWER THAN THE 4 WIRES:

- Top 4 Pins = VCC (Red Row 24), GND (Black Row 25), SCL (Yellow Row 26), SDA (Blue Row 27).
- 5th Bottom Pin = DRDY (Row 28, unused/bare). That 5th pin occupies 1 extra hole at the bottom!

RIGHT (GOLDEN-PATH PHOTO 56): 3 EMPTY HOLES IN COL f vs. 2 EMPTY IN COL j



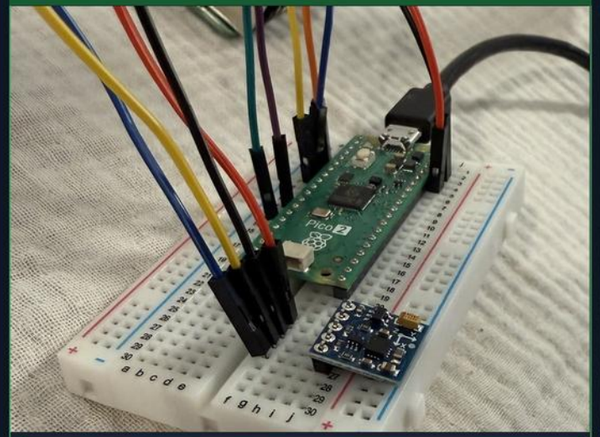
EXACT HOLE-COUNTING LANDMARKS (ZERO RELIANCE ON PRINTED NUMBERS):

- TOP CHECK: Leave 3 EMPTY HOLES below Pico 2 (Rows 21, 22, 23) in BOTH Col f and Col j.
- BOTTOM CHECK: 3 empty holes at bottom of Col f (f28-f30) | 2 empty holes at bottom of Col j (j29-j30)!

COMPLETED STAGE 2 GOLDEN-PATH BREADBOARD (PHOTO 57) & POWERED BENCH SYSTEM (PHOTO 58)

Left: Close-up of completed breadboard (Pico 2 + 7 wires + GY-271 + Micro-USB). Right: Full powered benchtop system (Photo 58).

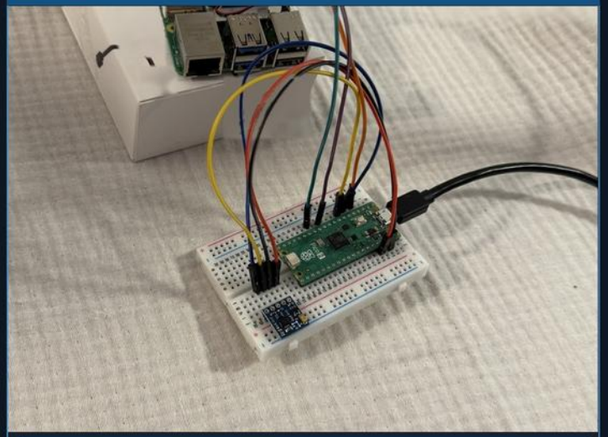
LEFT (PHOTO 57): COMPLETED BREADBOARD CLOSE-UP (PICO 2 + W1–W7 + COMPASS)



GOLDEN-PATH BREADBOARD WIRING — 100% VERIFIED:

- Pico 2 in Rows 1–20; W1–W4 (Red/Black/Yellow/Blue) in Col f Rows 24–27 (3 empty bottom holes).
- GY-271 5-Pin Compass in Col j Rows 24–28 (2 empty bottom holes) + Micro-USB in Pico 2 Row 1!

RIGHT (PHOTO 58): COMPLETE POWERED BENCHTOP SYSTEM (GOLDEN PATH)



STEP 2.5 POWER & USB CONNECTIONS — 100% VERIFIED:

- Cable 1: Black Micro-USB in Pico 2 (Row 1) -> Laptop USB for live serial telemetry.
- Cable 2: Official 27W USB-C Power Supply in Pi 5 'PWR IN' + Camera Module 3 on 200mm ribbon!

1. 58-PHOTO BUILD LOG

Real Hardware Verification

- Documented across 58 chronological benchtop photos (1 step per row).
- Covers Active Cooler push-pins, Pico 2 UF2 flashing, and MicroSD zero-metadata staging.

2. ZERO-SOLDER FIT

3 vs 2 Empty Hole Rule

- 15–30 deg pliers axial twist locks the 5-pin GY-271 header gas-tight.
- Landmark rule: 3 empty bottom holes in Col f vs 2 empty in Col j prevents any 1-row pin shift.

3. VIBRATION LOCKING

Snowmobile-Proof (<2% Risk)

- Keeps 100% generic COTS breadboard (zero custom PCB fabrication trail).
- Rigid backplate co-mount + 3M block binding + peelable Permatex RTV silicone damper bead.

4. VERIFIED STACK

Hyperfocal & 94 Tests

- Camera Module 3 NoIR Wide locked to hyperfocal focus (1m to infinity).
- Verified by 5 automated test suites + True 1:1 CAD vector drill sheet in the main build booklet.